

HIGH PERFORMANCE DATA PROCESSING
SECP3133
PROJECT 2

Vollzanbel

Real-Time Sentiment Analysis on Malaysian E-Wallet App Reviews

Built with Apache Kafka · Apache Spark · Elasticsearch · Kibana

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INTRODUCTION

The Problem & Why It Matters

1

Millions of daily users

E-wallets like Touch 'n Go, Boost and ShopeePay are part of everyday Malaysian life - payments, transfers, tolls, parking.

2

Opinion of the app

Users pour their experiences into Google Play reviews - a public, Malaysian-relevant signal of how these apps perform.

3

Too much to read manually

Reviews arrive continuously and in huge volume. Teams can't read them by hand or react in time.

OUR SOLUTION

A real-time pipeline that reads each review, classifies its sentiment and shows the trend live on a dashboard.

positive
neutral
negative

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INTRODUCTION

Objectives & Scope

OBJECTIVES

- Collect Malaysian-relevant review text
- Clean & preprocess with NLP techniques
- Train & compare two sentiment models
- Stream & classify in real time (Kafka + Spark)
- Store in Elasticsearch, visualize in Kibana
- Compare batch vs. streaming performance

SCOPE

APPS

9 Malaysian e-wallet / banking apps

CLASSES

Positive, Neutral and Negative

LANGUAGE

~97% English (Malay/Manglish handled)

MODELS

Logistic Regression + LSTM

STACK

Kafka → Spark → Elasticsearch → Kibana

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DATA ACQUISITION

9 Apps, 18,000 Google Play Reviews

18,000

raw reviews collected

2,000 reviews × 9 apps

Source: Google Play
Tool: google-play-scraper

THE APPS

- 1 Touch 'n Go eWallet
- 2 Maybank MAE
- 3 Boost eWallet
- 4 ShopeePay (Shopee MY)
- 5 BigPay
- 6 CIMB OCTO MY
- 7 RHB Banking App
- 8 Hong Leong Connect
- 9 MyPB by Public Bank

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PREPROCESSING & LABELLING

From Raw Text to Labelled Data

CLEANING PIPELINE

Handles English, Bahasa Malaysia & Manglish (Malay + English)

- Lowercase, emojis → words
- Remove URLs, emails, @mentions
- Expand contractions, collapse repeats
- Tokenize, remove stopwords (keep negations)
- Lemmatize with WordNet

LABELLING

Labels come from a pre-trained transformer reading the **review text**.

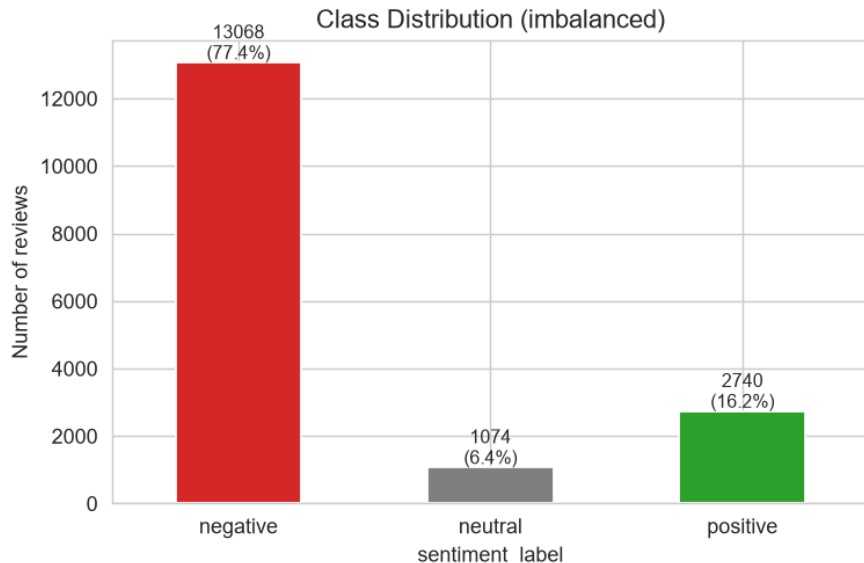
```
cardiffnlp/twitter-roberta-base-sentiment-latest
```

16,882 clean, labelled reviews

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THE DATA CHALLENGE

A Heavily Imbalanced Dataset



77.4% of reviews are negative

16.2% positive · 6.4% neutral

PROBLEM

A model that always says "negative" scores ~77% accuracy — but is useless.

Fix: stratified 70/20/10 split + balanced class weights + macro-F1 metric.

Two Models Compared

ML

Logistic Regression

- Machine Learning
- TF-IDF text features
- `class_weight = "balanced"`
- Lightweight & fast to serve

DL

LSTM

- Deep Learning (PyTorch)
- Word-embedding + LSTM layer
- Balanced weights in the loss function
- Heavier, captures word order

Both trained on the same stratified 70 / 20 / 10 split

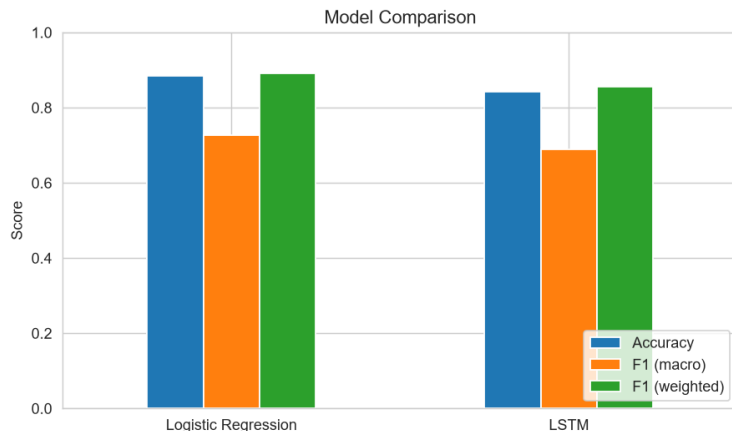
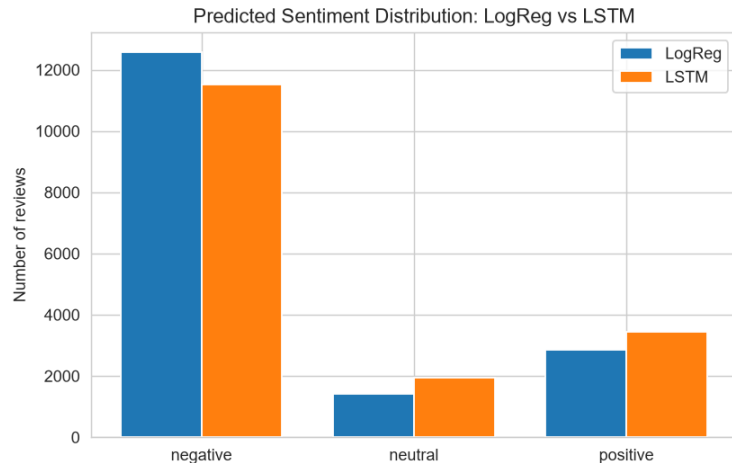
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MODEL EVALUATION

Logistic Regression Wins

Model	Accuracy	F1 (macro)	F1 (weighted)
Logistic Regression	0.885	0.726	0.891
LSTM	0.843	0.689	0.856

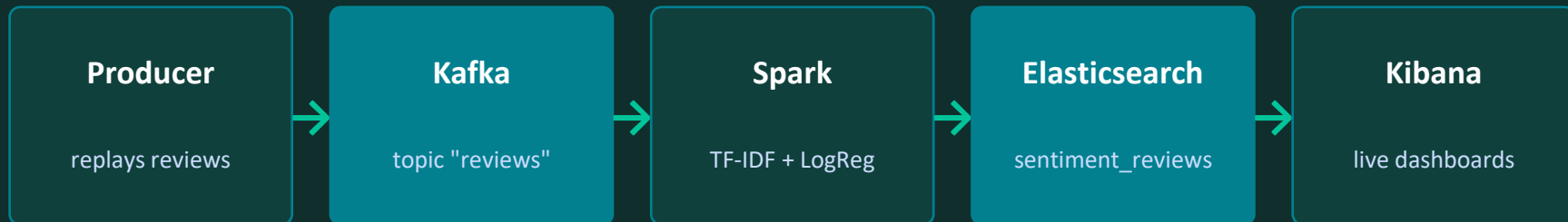
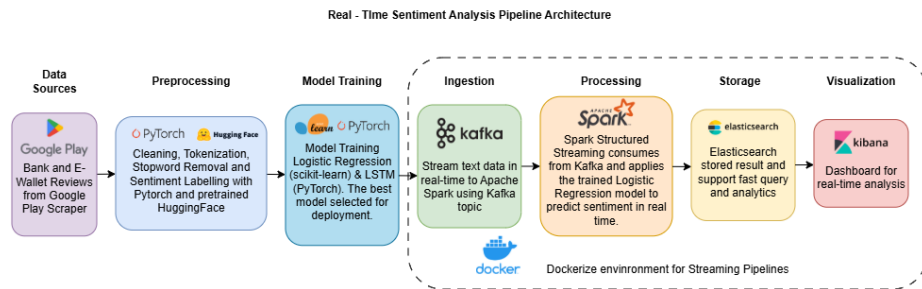
Logistic Regression beats the LSTM on every aggregate metric — and is far lighter to run per record, making it the right choice for a streaming workload.



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APACHE SYSTEM ARCHITECTURE

End-to-End Streaming Architecture



Orchestrated with Docker Compose

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ANALYSIS & RESULTS

Live Kibana Dashboard

1

Sentiment distribution

Pie chart - share of positive / neutral / negative

2

Sentiment over time

Line chart tracking sentiment as reviews arrive

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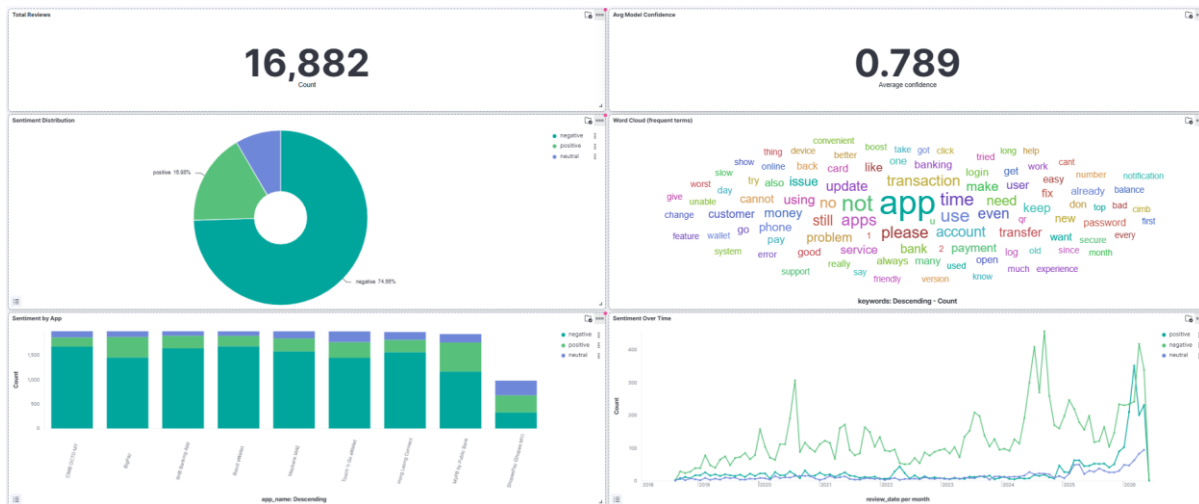
Per-app breakdown

Which apps draw the most negative reviews

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Real-time stream view

Live table, auto-refreshing as data flows in



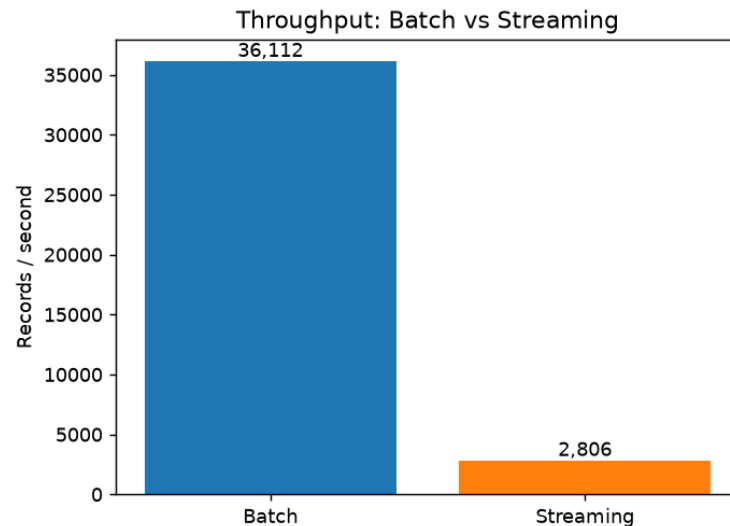
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OPTIMISATION & COMPARISON

Batch vs. Streaming

Metric	Batch	Streaming
Records processed	16,882	16,882
Throughput (rec/s)	~31,750	~2,806
Accuracy	0.942	0.948
Avg latency	—	~3.8 s

Batch wins raw speed. Streaming trades it for continuous, low-latency processing. **Accuracy is consistent (~0.94) across both.**



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KEY INSIGHTS

What the Data Tells Us

1

Malaysians are largely unhappy

77% of reviews are negative across all nine major apps. Reliability is the top concern.

2

Crashes & logins dominate complaints

Most frequent negative terms: app crashes, login/verification issues, removed features (auto top-up).

3

Real-time monitoring

The dashboard lets user to catch negative spikes the moment a bad release lands.

What We Built & What's Next

DELIVERED

- 18,000 reviews across 9 apps collected
- 16,882 text-labelled with a transformer
- Two models trained & compared
- Best model deployed in a live Kafka–Spark pipeline
- Results in Elasticsearch + Kibana
- Batch vs. streaming benchmarked

FUTURE WORK

- Add a fine-tuned BERT / DistilBERT model
- Stronger Bahasa Malaysia / Manglish handling
- Live continuous collection (not replay)
- Scale the Spark/Kafka cluster horizontally
- Add CPU / memory resource profiling

Thank You
